GATE - ALL BRANCHES

Engineering Mathematics

VECTOR SPACE

ONE SHOT







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Topics to be Covered



Topic

1. VECTOR SPACE
2. SUBSPACE
3. BASIS
4. DIMENSION of VECTOR SPACE
5. Concept of SPANNING in Vector Space.

VECTOR SPACE

Defn: → Any set of vectors V is called Vector space if it satisfies following conditions

1. if $x, y \in V \Rightarrow \boxed{x+y \in V}$ i.e. V is closed under addition (closure property)
2. if $x, y \in V$ then $\boxed{x+y=y+x}$ i.e. Commutative prop of V. addition.
3. if $x, y, z \in V$ then $\boxed{x+(y+z)=(x+y)+z}$ i.e. Associative prop of V. addition.
4. if $x \in V$ then $\exists\ 0 \in V$ s.t $\boxed{x+0=x}$ i.e. Additive identity exist
5. if $x \in V$ then $-x \in V$ i.e. $\boxed{x+(-x)=0}$ i.e. Additive inverse exist

⑥ if $X \in V$ & c is any scalar then $cX \in V$ i.e closure property for Multiplication

⑦ if $X \in V$ & λ, μ are any scalar then $\lambda(\mu X) = (\lambda\mu)X$ i.e Asso. prop of "

⑧ if $X \in V$ then $1.X \in V$ i.e Multiplicative identity exist.

⑨ if $X, Y \in V$ & λ is any scalar then $\lambda(X+Y) = \lambda X + \lambda Y$, Distributive Property
i.e Scalar Distributes over vector addition.

⑩ if $X \in V$ & λ, μ are scalars then $(\lambda+\mu)X = \lambda X + \mu X$, Distributive Property
i.e vector Distributes over Scalar Addition.

Any set of vectors satisfying above 10 properties is called V-Space.

Examples of Vector space →

$V = R^2, R^3, R^4 \ldots$ are the examples of V-space.

Let $V = R^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} ; x, y \in R \right\}$ = set of vectors.

① $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \in R^2, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \in R^2 \Rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} \in R^2$ ie closure prop for addition holds.

② $\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ ie Commutative prop for V. Addition Holds

③ $\because \underbrace{\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix}} + \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} + \begin{bmatrix} 4 \\ -3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ ie Associative prop for V. Addition Holds.

while $\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \underbrace{\begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ -3 \end{bmatrix}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$

④ $\because \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in R^2 \Rightarrow \begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \in R^2 \Rightarrow$ Additive identity exist.

⑤ $\because \begin{bmatrix}-2\\-3\end{bmatrix} \in R^2$ i.e. $\begin{bmatrix}2\\3\end{bmatrix}+\begin{bmatrix}-2\\-3\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix} \Rightarrow$ Additive inverse exist

⑥ $\because \begin{bmatrix}2\\3\end{bmatrix} \in R^2$ & let 5 is any scalar then $5\begin{bmatrix}2\\3\end{bmatrix}=\begin{bmatrix}10\\15\end{bmatrix} \in R^2$ i.e. $\boxed{cx \in R^2}$

i.e. Closure prop for Multiplication Holds.

⑦ $\because \begin{bmatrix}2\\3\end{bmatrix} \in R^2$, 3 & 5 are two scalars then $(3\times 5)\begin{bmatrix}2\\3\end{bmatrix}=15\begin{bmatrix}2\\3\end{bmatrix}=\begin{bmatrix}30\\45\end{bmatrix}$

& Associative prop for Multi Holds. & $3\{5\begin{bmatrix}2\\3\end{bmatrix}\}=3\begin{bmatrix}10\\15\end{bmatrix}=\begin{bmatrix}30\\45\end{bmatrix}$

⑧ $\because 1.\begin{bmatrix}2\\3\end{bmatrix}=\begin{bmatrix}2\\3\end{bmatrix} \in R^2$ i.e. Multi Identity exist.

⑨ $\because (2+3)\begin{bmatrix}2\\3\end{bmatrix}=\begin{bmatrix}10\\15\end{bmatrix}=2\begin{bmatrix}2\\3\end{bmatrix}+3\begin{bmatrix}2\\3\end{bmatrix}$ i.e. Dist. prop Holds.

⑩ $5\left[\begin{bmatrix}2\\3\end{bmatrix}+\begin{bmatrix}-1\\2\end{bmatrix}\right]=5\begin{bmatrix}1\\5\end{bmatrix}=\begin{bmatrix}5\\25\end{bmatrix}=5\begin{bmatrix}2\\3\end{bmatrix}+5\begin{bmatrix}-1\\2\end{bmatrix}$ i.e. Dist. Prop Holds.

Subspace → Let V is any V.space & let $W \subset V$ (i.e W is a subset of V) then

→ W is called subspace if it is itself a vector space.

→ W is called subspace if it also satisfy all 10 properties of V-space.

Shortcut → W is called subspace of any vector space V if.

$\forall\, x, y \in W \Rightarrow$

- $x+y \in W$ ie W is closed under vector addition
- $cx \in W$ ie W is also closed under scalar Multi

OR

$\forall\, x, y \in W$ & c is any scalar $\Rightarrow cx+y \in W$ then W is called subspace.

SPANNING in Vector space → Let W is subspace of any vector space V.

Let S is any subset of W ie $S \subset W$

Linear Span: set of all linear combinations of vectors (that lies in S) is called linear span of S and it is denoted as $L(S)$

eg Let $S = \{x_1, x_2, x_3 \dots x_r\}$ then $L\{S\} = \{(c_1x_1 + c_2x_2 + \dots + c_rx_r);\ c_i = \text{scalars},\ x_i = \text{vectors}\}$

Spanning: if $L\{S\} = W$ then W is spanned by S (generated) or S spans W (generates)

ie "Any Random Vector in W can be generated by linear combinations of vectors in S"

ie Different-Different linear comb of vectors in S provides diff-diff vectors of W

Let us take an example, $S=\{x_1,x_2\}=\left\{\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right\}$ where $x_1,x_2\in R^2$ i.e $S\subset R^2$

then S spans R^2 i.e $L(S)=R^2$

Let $X=\begin{bmatrix}2\\-3\end{bmatrix}\in R^2$ then $\begin{bmatrix}2\\-3\end{bmatrix}=\begin{bmatrix}2\\0\end{bmatrix}+\begin{bmatrix}0\\-3\end{bmatrix}=2\begin{bmatrix}1\\0\end{bmatrix}-3\begin{bmatrix}0\\1\end{bmatrix}$

$X=2x_1-3x_2$

Let $Y=\begin{bmatrix}\lambda\\\mu\end{bmatrix}\in R^2$ then $Y=\begin{bmatrix}\lambda\\\mu\end{bmatrix}=\begin{bmatrix}\lambda\\0\end{bmatrix}+\begin{bmatrix}0\\\mu\end{bmatrix}=\lambda\begin{bmatrix}1\\0\end{bmatrix}+\mu\begin{bmatrix}0\\1\end{bmatrix}=\lambda x_1+\mu x_2$

i.e we have justified that Any Random Member of R^2 can be expressed as a Linear Combination of x_1 & x_2 i.e $L\{S\}=R^2\Rightarrow S$ generates R^2 or S Spans R^2

Similarly if we take $S=\{x_1 x_2 x_3\}=\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}0\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}=\{x_1,x_2,x_3\}$ i.e $S\subset R^3$

then S will generate R^3 i.e **S Spans R^3** or $L\{S\}=R^3$

Cross check, Let $X=\begin{bmatrix}\alpha\\\beta\\\gamma\end{bmatrix}\in R^3$ then $X=\begin{bmatrix}\alpha\\\beta\\\gamma\end{bmatrix}=\begin{bmatrix}\alpha\\0\\0\end{bmatrix}+\begin{bmatrix}0\\\beta\\0\end{bmatrix}+\begin{bmatrix}0\\0\\\gamma\end{bmatrix}$

$$X=\alpha\begin{bmatrix}1\\0\\0\end{bmatrix}+\beta\begin{bmatrix}0\\1\\0\end{bmatrix}+\gamma\begin{bmatrix}0\\0\\1\end{bmatrix}$$

$$X=\alpha X_1+\beta X_2+\gamma X_3$$

i.e any Random Member X of R^3 can be expressed as a L-Comb of X_1, X_2, X_3

$\Rightarrow S$ Spans $R^3 \Rightarrow L\{S\}=R^3$

Ex to understand BASIS of any V.Space

R^2 $\begin{cases} x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \end{bmatrix} \text{---} \text{ on x axis} \\ x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \end{bmatrix} \text{---} \text{ on y axis} \\ \longrightarrow X = c_1 x_1 + c_2 x_2 \quad (\text{elements of I, II, III, IV Quad.}) \end{cases}$

So to form R^2 we need two L.I set of vectors that generates R^2.
this New set is called BASIS of R^2 (spans)

$S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix} \right\} = \{x_1, x_2, x_3, x_4\}$ then S spans R^2 but S is not L.I

BASIS of Vector space → A set of vectors S is called Basis for V.space V if ① S spans V ② S must consist L.I vectors.

eg: ∵ $V=R^2$ is vector space then $S=\{x_1,x_2\}=\left\{\begin{bmatrix}1\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right\}$ is called Basis for R^2 ∵
- (i) S spans R^2
- (ii) x_1 & x_2 are LI (By observation)

eg: ∵ $V=R^3$ is vector space then $S=\{x_1x_2x_3\}=\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}0\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ is called Basis for R^3 ∵
- (i) S spans R^3
- (ii) x_1,x_2,x_3 are LI $(\because |S|\neq 0)$

RECAP: Let $X_1, X_2, X_3 \ldots X_r$ are the given set of vectors

& Consider $A = [X_1\ X_2\ X_3 \ldots X_r]$

① General Method

(i) if $S(A)$ = No. of vectors $\Rightarrow$ LI

(ii) if $S(A) <$ " " $\Rightarrow$ LD

② Tricky Method (if A is Sq. Mat)

(i) if $|A| \neq 0 \Rightarrow$ vectors are LI

(ii) if $|A| = 0 \Rightarrow$ " " LD

Note: In Case of two vectors X_1 & X_2, there is no Need to follow G. Method or Tricky Method, only follow Observation Method.

Dimension of vector space →

dim (V.Space) = Counting of vectors in it's Basis

for eg, $\dim(R^2) = \text{two}$ ∵ $S = \{x_1, x_2\} = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$ contains two vectors

$\dim(R^3) = \text{three}$ ∵ $S = \{x_1 x_2 x_3\} = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$,, three vectors

Let W is subspace of R^3 then W is also a V.Space Hence we have following three cases: ① dim (W) may be 3, ② dim (W) may be 2 ③ dim (W) maybe 1.

Although vectors of W are 3-dimensional vectors.

∮ for $V = R^2$, Basis is $S = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$ But X axis is subspace of R^2 ie $X = \left\{\begin{bmatrix}a\\0\end{bmatrix}, a \in R\right\}$

& Basis for X axis is $S = \left\{\begin{bmatrix}1\\0\end{bmatrix}\right\}$ so dim (X) = one.

PROPERTIES of BASIS & DIMENSION →

① Basis of any vector space is **Not unique**

② if $\dim(V) = n$ then Any set of n, LI vectors forms Basis.

eg :: $\dim(R^3) = 3$ then any set of 3, LI vectors will be Basis for R^3

eg :: $\dim(R^4) = 4$ then any set of 4, LI vectors will form Basis for R^4.

Q: Prove that $S = \{(1\,2\,1)^t, (2\,1\,-4)^t, (3\,-2\,1)^t\}$ is also Basis for R^3

Note: Standard Basis for $R^3 = \{\epsilon_1, \epsilon_2, \epsilon_3\} = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$

Sol: $S = \{x_1, x_2, x_3\} = \left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}2\\1\\-4\end{bmatrix}, \begin{bmatrix}3\\-2\\1\end{bmatrix}\right\}$ $\because |S| = \cdots\cdots = -42$ (HW)

$\because |S| \neq 0$ so x_1, x_2, x_3 are LI (2nd condition for Basis Holds)

Now for spanning, Consider any Random Vector of R^3 say $X = \begin{bmatrix}2\\1\\3\end{bmatrix}$

Let us Consider $k_1x_1 + k_2x_2 + k_3x_3 = X$

$$k_1\begin{bmatrix}1\\2\\1\end{bmatrix} + k_2\begin{bmatrix}2\\1\\-4\end{bmatrix} + k_3\begin{bmatrix}3\\-2\\1\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix} \quad (1)$$

$$k_1 + 2k_2 + 3k_3 = 2$$
$$2k_1 + k_2 - 2k_3 = 1$$
$$k_1 - 4k_2 + k_3 = 3$$

$$\begin{bmatrix}1 & 2 & 3\\2 & 1 & -2\\1 & -4 & 1\end{bmatrix}\begin{bmatrix}k_1\\k_2\\k_3\end{bmatrix} = \begin{bmatrix}2\\1\\3\end{bmatrix} \Rightarrow \boxed{SX = B}$$

$\because |S| \neq 0$ so By Matrix Method, unique Sol. will exist. i.e unique Values of k_1, k_2, k_3 exist.

i.e. $X = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}$ = unique values exist. i.e. eqn^n (1) is justified.

$\Rightarrow x_1, x_2, x_3$ together generates X i.e. S generates $R^3 \Rightarrow$ **S spans R^3**

$\because$ S contains LI set of vectors & S spans $R^3 \Rightarrow S$ is also a Basis for R^3.

Conclusion: BASIS is not unique for any V. Space.

(2) Here we have also verified that,

$\dim(R^3) = 3 \Rightarrow$ Any set of 3 LI vectors forms Basis of R^3.

(Ex): Standard Basis for R^4 is $S_1 = \{e_1, e_2, e_3, e_4\} = \left\{\begin{bmatrix}1\\0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\0\\1\end{bmatrix}\right\}$

& $\dim(R^4) = ? = 4$ (∵ Basis contains 4 vectors)

(Ex) P.T. $S_2 = \{x_1, x_2, x_3, x_4\} = \left\{\begin{bmatrix}0\\1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\0\\3\\0\end{bmatrix}, \begin{bmatrix}2\\3\\0\\1\end{bmatrix}, \begin{bmatrix}3\\0\\1\\2\end{bmatrix}\right\}$ forms Basis for R^4.

$$S_2 = \begin{bmatrix}0&1&2&3\\1&0&3&0\\2&3&0&1\\3&0&1&2\end{bmatrix}_{4\times4}$$

∵ $|S_2| = \cdots\cdots\cdots = 88$ (HW)

i.e. $|S_2| \neq 0 \Rightarrow x_1, x_2, x_3, x_4$ are (LI).

∴ $\dim(R^4)$ is also 4 so these vectors will form Basis for R^4 (Using prop 2).

eg (PT) $S_3 = \{\gamma_1\ \gamma_2\ \gamma_3\ \gamma_4\} = \left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}\right\}$ is not Basis for R^4.

(M-I) $\because \gamma_4 = \gamma_1 + \gamma_2 + \gamma_3$ i.e. these 4 vectors are LD vectors
so they will not form Basis.

(M-II) $\because |S_3| = \begin{vmatrix}1&0&0&1\\0&1&0&1\\0&0&1&1\\0&0&0&0\end{vmatrix} = 0$ i.e. vectors are (LD)

Q: Consider the set of column vectors defined by $X=\{(x_1\ x_2\ x_3)^T;\ x_1+x_2+x_3=0\}$ then $S=\{(1\ -1\ 0)^T, (1\ 0\ -1)^T\}$ is Basis for subspace X ?

Sol: Here X is subspace of R^3 i.e $X \subset R^3$

$S=\{x_1, x_2\}=\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\-1\end{bmatrix}\right\}$ Here x_1 & x_2 are (LI) (By observation)

Now we will try to check whether $L(S)=X$?? i.e S generates X ??

Let us take Random member of X say $Y=\begin{bmatrix}2\\-3\\1\end{bmatrix}$ then

$$Y=\begin{bmatrix}2\\-3\\1\end{bmatrix}=3x_1-x_2=3\begin{bmatrix}1\\-1\\0\end{bmatrix}-\begin{bmatrix}1\\0\\-1\end{bmatrix}=\begin{bmatrix}3\\-3\\0\end{bmatrix}+\begin{bmatrix}-1\\0\\1\end{bmatrix}=\begin{bmatrix}2\\-3\\1\end{bmatrix}$$ i.e S spans X

Hence S is Basis for subspace X & dim(X) = No. of vectors in Basis = (TWO)

Analysis :- $V = R^3 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} ; x_1 \in R, x_2 \in R, x_3 \in R \right\}$ = 3-D Space.

is V is a vector space.

Let us take $X = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} ; x_1 + x_2 + x_3 = 0 \right\}$ obviously $X \subset R^3$

$\because$ X is also satisfying all the 10 properties of V. Space

So X is also a vector space & it (X) is called SubSpace of R^3

& $\dim(R^3) = 3$ But $\dim(X) = 2$

Consider the set of (column) vectors defined by $X = \{x \in R^3 \mid x_1 + x_2 + x_3 = 0$, where $x^T = [x_1, x_2, x_3]^T\}$. Which of the following is TRUE?

(a) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a basis for the subspace X. (T)

(b) $\{[1, -1, 0]^T, [1, 0, -1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X (f)

(c) X is not a subspace for R^3 (f) (see previous slide)

(d) None of the above

If the vectors, $e_1 = (1,0,2), e_2 = (0,1,0)$ and $e_3 = (-2,0,1)$ from an orthogonal basis of the three dimensional real space $\mathbb{R}^3$, then the vector $u = (4,3,-3) \in \mathbb{R}^3$ can be expressed as

(a) $u = -\frac{2}{5}e_1 - 3e_2 - \frac{11}{5}e_3$

(b) $u = -\frac{2}{5}e_1 - 3e_2 + \frac{11}{5}e_3$

(c) $u = -\frac{2}{5}e_1 + 3e_2 + \frac{11}{5}e_3$

(d) $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

$u = \begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix}$, Basis $S = \{e_1 e_2 e_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} \right\}$

S will generates R^3 i.e. S Spans R^3

Any Member of R^3 can be expressed as a linear comb of vectors in S.

$k_1e_1 + k_2e_2 + k_3e_3 = u$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix} \Rightarrow X = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 3 \\ -11/5 \end{bmatrix}$$

$SX = u$

Ans (d)

(a) $4 = -\frac{2}{5}(1) - 3(0) - \frac{11}{5}(-2) = 4$

$3 = \frac{2}{5}(0) - 3(1) - \frac{11}{5}(0) \neq 3$

(b) X (c) X (d) ✓

$$\begin{bmatrix}4\\3\\-3\end{bmatrix} = k_1\begin{bmatrix}1\\0\\2\end{bmatrix} + k_2\begin{bmatrix}0\\1\\0\end{bmatrix} + k_3\begin{bmatrix}-2\\0\\1\end{bmatrix}$$

$$k_1\begin{bmatrix}1\\0\\2\end{bmatrix} + k_2\begin{bmatrix}0\\1\\0\end{bmatrix} + k_3\begin{bmatrix}-2\\0\\1\end{bmatrix} = \begin{bmatrix}4\\3\\-3\end{bmatrix}$$

$$\begin{cases} k_1 + 0k_2 - 2k_3 = 4 \\ 0k_1 + k_2 + 0k_3 = 3 \\ 2k_1 + 0k_2 + k_3 = -3 \end{cases}$$

$$\begin{bmatrix}1 & 0 & -2\\0 & 1 & 0\\2 & 0 & 1\end{bmatrix}\begin{bmatrix}k_1\\k_2\\k_3\end{bmatrix} = \begin{bmatrix}4\\3\\-3\end{bmatrix}$$

$AX = B$.

$\because |A| = \ldots = 5$ i.e. $|A| \neq 0$

so unique sol exist.

$$[A:B] = \begin{bmatrix}1 & 0 & -2 & : & 4\\0 & 1 & 0 & : & 3\\2 & 0 & 1 & : & -3\end{bmatrix} \rightarrow \begin{bmatrix}1 & 0 & -2 & : & 4\\0 & 1 & 0 & : & 3\\0 & 0 & 5 & : & -11\end{bmatrix}$$

$$\begin{bmatrix}1 & 0 & -2\\0 & 1 & 0\\0 & 0 & 5\end{bmatrix}\begin{bmatrix}k_1\\k_2\\k_3\end{bmatrix} = \begin{bmatrix}4\\3\\-11\end{bmatrix}$$

$$\left.\begin{aligned} 5k_3 &= -11 \\ k_2 &= 3 \\ k_1 - 2k_3 &= 4 \end{aligned}\right\} \quad \begin{aligned} k_3 &= -11/5 \\ k_2 &= 3 \\ k_1 &= 4 - \frac{22}{5} = -\frac{2}{5} \end{aligned}$$

By (1) $u = -\frac{2}{5}e_1 + 3e_2 - \frac{11}{5}e_3$

It is given that $X_1, X_2, \ldots X_M$ are M non-zero, orthogonal vectors. The dimension of the vector space spanned by the 2M vectors $X_1, X_2, \ldots X_M, -X_1, -X_2, \ldots -X_M$ is

(a) 2M

(b) M + 1

(c) M

(d) dependent on the choice of $X_1, X_2, \ldots X_M$

$L(S) = \{X_1, X_2, \ldots X_M, -X_1, -X_2, \ldots, -X_M\}$

Basis $= \{X_1, X_2, X_3 \ldots X_M\}$ → orthogonal vectors ⇓ LI vectors.

$\therefore \dim = M$.

LD. ∵ $\dim(R^4) = 4$.

If $v_1, v_2, \ldots, v_6$ are six vectors in R^4, which one of the following statements is False?

(a) If $\{v_1, v_3, v_5, v_6\}$ spans R^4, then it forms a basis for R^4. (T) ∵ $\dim(R^4) = 4$, so Minimal spanning set contain 4 vectors.

(b) These vectors are not linearly independent. (T)

(c) It is not necessary that these vectors span R^4. (True)

(d) Any four of these vectors form a basis for R^4. (false) If we are taking all 6 vectors from single family then they will not span R^4 Hence this option is False.

Q) Let M_n denote the vector space of all $n \times n$ real matrices. Consider the following subsets of M_n

MSQ

1. $W_1 = \{A \in M_n \mid A^2 = I\}$, where I is the identity matrix.
2. $W_2 = \{A \in M_n \mid \text{rank}(A) = 1\}$
3. $W_3 = \{A \in M_n \mid \text{trace}(A) = 0\}$
4. $W_4 = \{A+B \mid A \in M_n, B \text{ is a fixed matrix in } M_n\}$

Which of the following statements is correct?

(a) W_1 is a linear subspace of M_n ✗

(b) W_2 is a linear subspace of M_n ✗

(c) ✓ W_3 is a linear subspace of M_n (T)

(d) ✓ W_4 is a linear subspace of M_n only if B is a zero matrix.

(b) $W_2 = \{A \in M_n ; \rho(A) = 1\}$

Let $A, B \in M_n$ then $\rho(A) = 1$ & $\rho(B) = 1$

Now $\rho(A+B) \neq \rho(A) + \rho(B)$

$\therefore \rho(A+B) \leq \rho(A) + \rho(B)$

i.e. $A+B \notin W_2$

So W_2 is not subspace.

(a) $W_1 = \{A \in M_n : A^2 = I\}$, Let $A, B \in W_1$ then $A^2 = I$ & $B^2 = I$ (given)

$(A+B)^2 = A^2 + AB + BA + B^2 = I + AB + BA + I$

$\neq I$ so $(A+B) \notin W_1$ so W_1 is not subspace.

© $W_3 = \{A \in M_n : Tr(A) = 0\}$

Let $A, B \in M_n \Rightarrow Tr(A) = 0 = Tr(B)$

Now $Tr(A+B) = Tr(A) + Tr(B) = 0 + 0 = 0$

i.e. $A+B \in M_n$ Hence

M_n is closed under Vector Addition.

Now $Tr(cA) = c \cdot Tr(A) = c(0) = 0$

i.e. $cA \in M_n$ Hence

M_n is closed under Scalar Multiplication.

So M_n is a Vector space or subspace.

(d) $W_4 = \{A+B : A \in M_n \text{ & } B = \text{fixed Mat}\}$

We have two possibilities,

(i) Either B = Zero Mat.

then W_4 has additive identity.

then W_4 will become subspace.

(ii) OR $B \neq$ ZERO Mat.

then W_4 does not have additive identity

So W_4 will not be a subspace

i.e. (d) ✓

$P = \begin{bmatrix} -10 \\ -1 \\ 3 \end{bmatrix}^T$, $Q = \begin{bmatrix} -2 \\ -5 \\ 9 \end{bmatrix}^T$ and $R = \begin{bmatrix} 2 \\ -7 \\ 12 \end{bmatrix}^T$ are three vectors. An orthogonal set of vectors having a span that contains P, Q, R is

(a) $\begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix} \quad \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}$ $\Rightarrow x_1 \cdot x_2 = 0$

(b) $\begin{bmatrix} -4 \\ 2 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 5 \\ 7 \\ -11 \end{bmatrix} \quad \begin{bmatrix} 8 \\ 2 \\ -3 \end{bmatrix}$ $\Rightarrow$ Not orthogonal.

(c) $\begin{bmatrix} 6 \\ 7 \\ -1 \end{bmatrix} \quad \begin{bmatrix} -3 \\ 2 \\ -2 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 9 \\ -4 \end{bmatrix}$ $\Rightarrow$ " "

(d) $\begin{bmatrix} 4 \\ 3 \\ 11 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 31 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix}$ $\Rightarrow$ " "

M-I

M-II HW.

The following vector is linearly dependent upon the solution to the previous problem

HW

(a) $\begin{bmatrix} 8 \\ 9 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} -2 \\ -17 \\ 30 \end{bmatrix}$

(c) $\begin{bmatrix} 4 \\ 4 \\ 5 \end{bmatrix}$

(d) $\begin{bmatrix} 13 \\ 2 \\ -3 \end{bmatrix}$

Q) Let S_1, S_2, S_3 be sets of real-valued functions defined as: (HW)

$S_1 = \{ f \mid f(3) = 0 \}$

$S_2 = \{ g \mid g(x) = x+1 \text{ for all } x \in R \}$

$S_3 = \{ h \mid h(x) = C \text{ for some instant } C \in K \}$

Which of the following statements is correct?

a) S_1 is a vector space & S_2 is not a vector space.

b) S_2 is a vector space & S_3 is not a vector space.

c) S_1 & S_3 are vector spaces, but S_2 is not a vector space

d) S_2 & S_3 are vector spaces but S_1 is not a vector space.

Solutions of Home Work Questions

(See Next Slides)

$P = \begin{bmatrix} -10 \\ -1 \\ 3 \end{bmatrix}^T$, $Q = \begin{bmatrix} -2 \\ -5 \\ 9 \end{bmatrix}^T$ and $R = \begin{bmatrix} 2 \\ -7 \\ 12 \end{bmatrix}^T$ are three vectors. An orthogonal set of vectors having a span that contains P, Q, R is

(a) $\begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix}$ $\begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}$ $X_1 \cdot X_2 = 0$ ✓ orthogonal

(b) $\begin{bmatrix} -4 \\ 2 \\ 4 \end{bmatrix}$ $\begin{bmatrix} 5 \\ 7 \\ -11 \end{bmatrix}$ $\begin{bmatrix} 8 \\ 2 \\ -3 \end{bmatrix}$ Not orthogonal

(c) $\begin{bmatrix} 6 \\ 7 \\ -1 \end{bmatrix}$ $\begin{bmatrix} -3 \\ 2 \\ -2 \end{bmatrix}$ $\begin{bmatrix} 3 \\ 9 \\ -4 \end{bmatrix}$ " "

(d) $\begin{bmatrix} 4 \\ 3 \\ 11 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 31 \\ 3 \end{bmatrix}$ $\begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix}$ " "

M-II As $P, Q, R \in L\{S\}$

Let us take (a) i.e.

$S = \{X_1, X_2\} = \left\{ \begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix} \right\}$

$P = X_1 - X_2$, $Q = X_1 + X_2$

& $R = X_1 + 2X_2$

i.e. $P, Q, R \in L\{S\}$ where S = option (a) ✓

The following vector is linearly dependent upon the solution to the previous problem

(a) $\begin{bmatrix} 8 \\ 9 \\ 3 \end{bmatrix} = x_3$

(b) $\begin{bmatrix} -2 \\ -17 \\ 30 \end{bmatrix} = x_4$

(c) $\begin{bmatrix} 4 \\ 4 \\ 5 \end{bmatrix} = x_5$

(d) $\begin{bmatrix} 13 \\ 2 \\ -3 \end{bmatrix} = x_6$

$S = \{x_1 x_2\} = \left\{\begin{bmatrix} -6 \\ -3 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 3 \end{bmatrix}\right\}$

$S_a = \{x_1 x_2 x_3\} \Rightarrow |S_a| \neq 0$ ie (LI)

$S_b = \{x_1 x_2 x_4\} \Rightarrow |S_b| = 0$ ie (LD) ✓

M-II : In option (b),

$x_4 = 3x_1 + 4x_2$

$= \begin{bmatrix} -18 \\ -9 \\ 18 \end{bmatrix} + \begin{bmatrix} 16 \\ -8 \\ 12 \end{bmatrix}$ ✓

Q) Let S_1, S_2, S_3 be sets of real-valued functions defined as:

$S_1 = \{ f \mid f(3) = 0 \}$

$S_2 = \{ g \mid g(x) = x+1 \text{ for all } x \in R \}$

$S_3 = \{ h \mid h(x) = C \text{ for some instant } C \}$

Which of the following statements is correct?

a) S_1 is a vector space & S_2 is not a vector space.

b) S_2 is a vector space & S_3 is not a vector space ✗

c) ✓ S_1 & S_3 are vector spaces, but S_2 is not a vector space

d) S_2 & S_3 are vector spaces but S_1 is not a vector space. ✗

$S_3 = \{ h(x) : h(x) = C \}$

Let $h, g \in S_3$ i.e. $h(x) = C$ & $g(x) = C$

$(h+g)(x) = h(x) + g(x) = C + C$

$= 2C = C_1 \Rightarrow h+g \in S_3$

$(\alpha h)(x) = \alpha(h(x)) = \alpha(C) = C_2$

$\Rightarrow \alpha h \in S_3$

So S_3 is also V. Space.

$S_1 = \{ f(x) : f(3) = 0 \}$

Let f & $g \in S_1$ i.e. $f(3) = 0, g(3) = 0$

Now $(f+g)(x) = f(x) + g(x) = 0 + 0 = 0$ i.e. $f+g \in S_1$

& $(cf)(x) = c\,f(x) = c(0) = 0$ i.e. $cf \in S_1$

$\Rightarrow S_1$ is a V. Space

Let $S_2 = \{ g(x) : g(x) = x+1 \}$

Let $g_1(x), g_2(x) \in S_2 \Rightarrow g_1(x) = x+1$ & $g_2(x) = x+1$

$$(g_1+g_2)(x) = g_1(x) + g_2(x)$$

$$= (x+1)+(x+1) = 2x+2 = 2(x+1)$$

$$\neq (x+1) \text{ i.e } g_1+g_2 \notin S_2$$

$\therefore S_2$ is Not V. Space.

THANK - YOU